Bridging realities: effectiveness of ai and immersive technologies in cross- platform next-gen museum experiences for storytelling and interaction

Project ID: 25-26J-416

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DEPLOYMENT REPORT

04/04/2026

Executive Summary

Bridging Realities is an immersive museum platform that uses augmented and virtual reality together with AI-driven narration and interaction. It is designed to move visitors beyond static labels and displays toward active exploration of fossils, geological time periods, and prehistoric environments, while supporting different languages, learning levels, and accessibility needs.

The platform is organized into four complementary experiences: two focus on **VR** (a fossil excavation simulation with tools, layers, achievements, and multiplayer excavation; and a multiplayer virtual museum with galleries, a jungle simulation, and AI-assisted voice interaction), and two focus on **mobile AR** (an exhibit guide that uses QR codes, 3D models, and adaptive AI voice guidance; and a geological-history experience that links coloring activities to AR, quizzes, and era-based storytelling). Shared technologies include Unity, Vuforia and AR Foundation where applicable, OpenXR for VR, Google Cloud Text-to-Speech and LLM services for intelligent narration, and Photon for multiplayer VR.

Deployment is structured as **two applications**: one **VR application** for Meta Quest 2 and 3, combining both VR experiences into a single install; and one **AR application** for smartphones, combining both AR experiences into a single install. AI and networking services may run in the cloud or locally depending on configuration, allowing use in museums and similar venues with varying infrastructure.

Overall, **Bridging Realities** targets higher engagement, clearer learning outcomes, and broader accessibility than traditional exhibit-only visits, with a path for pilot deployment, feedback, and iterative improvement in real museum settings.

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1. Project context

Focus of Bridging Realities

Bridging Realities targets museums and informal learning spaces where exhibits often rely on static text and limited interaction. The project asks how immersive technologies can improve how visitors stay engaged, remember what they learn, and access content across ages, languages, and abilities compared with traditional displays alone.

Research focus

The guiding question is whether immersive technologies: VR, AR, and mixed setups, can measurably strengthen visitor engagement, learning outcomes, and educational accessibility in museum settings when compared with conventional static exhibits.

Stakeholders

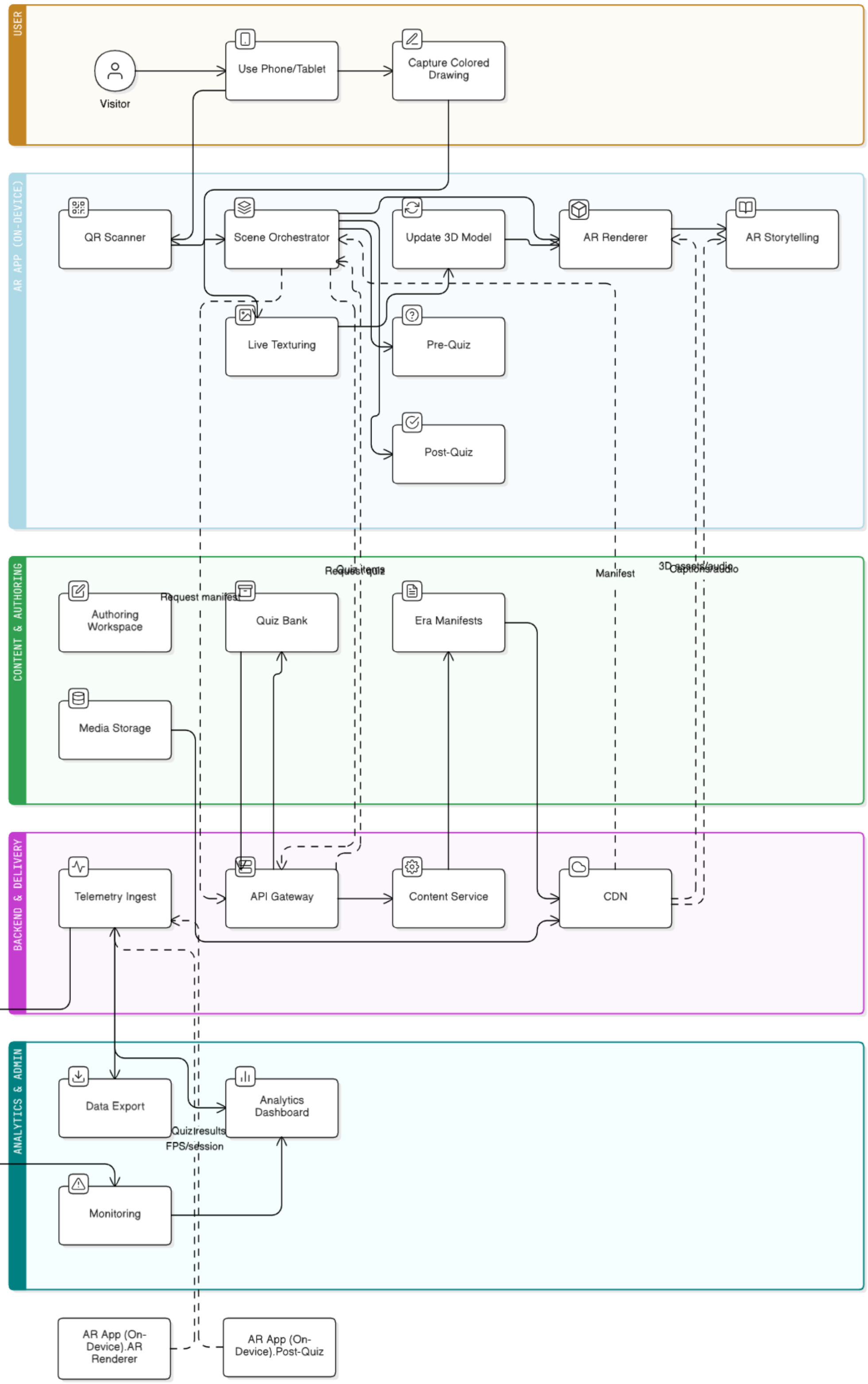
- Visitors such as students, families, and researchers are the primary users who experience the platform directly.
- Museum educators and curators shape how content is used in visits and programs.
- Museum administrators influence adoption, budget, and long-term operation.

Quality and constraints

The work emphasizes usability, smooth real-time graphics, accessibility for diverse users, and scalability so the same approach can suit small and large venues without a complete redesign of the technical approach.

2. System overview (four components)

2.1 Overall System Diagram



The solution comprises four components (two VR, two AR), each with defined features and technology stack per project documentation.

2.2 Component 1: Virtual Fossil Excavation Adventure (VR)

Aspect	**Description**
Owner	Ashen Tennakoon (IT22361004)
Role	Hands-on paleontology learning in VR
Features	Tool selection (brush, pick, chisel); stratified excavation through geological layers; achievement system (badges for techniques and species); multiplayer mode (collaborate or compete)
Technologies	Unity 3D with **OpenXR**; **Meta Quest 3**; realistic soil deformation and artifact interaction physics; AI tutoring for real-time guidance and feedback

2.3 Component 2: Multiplayer VR Museum with Jungle Simulation (VR)

Aspect	Description
Owner	Sherwin Li (IT22135520)
Role	Shared exploration of museum content and prehistoric environments
Features	Virtual museum galleries (fossil exhibits, information panels); jungle simulation with prehistoric environments and animated creatures; multiplayer avatars; voice commands with AI-generated explanations
Technologies	Unity 3D with OpenXR; Photon PUN 2 (multiplayer); Meta Quest 2 / 3; LLM-based AI voice assistant for natural language interaction

2.4 Component 3: Interactive AR Mobile Guide with AI Voice Narration (AR)

Aspect	**Description**
Owner	Imalki Pinto (IT22149558)
Role	Turn static exhibits into interactive, multilingual learning
Features	QR-triggered AR for 3D fossil models; adaptive narration (language and level); gamification (quizzes, badges, challenges); accessibility (adjustable text, speech speed, color-blind friendly UI)
Technologies	**Unity AR Foundation** + **Vuforia**; **Android (ARCore)** and **iOS (ARKit)**; photogrammetry-based 3D assets; **Google Cloud Text-to-Speech**; **LLM-based** narration

2.5 Component 4: Interactive AR Exploration of Earth's Geological History (AR)

Aspect	**Description**
Owner	Ruwani Pradeepa (IT22033482)
Role	Connect coloring activities with AR storytelling for geological learning
Features	Coloring-to-AR (drawings animate in AR); interactive quizzes (adaptive to performance); narrative exploration (Paleozoic, Mesozoic, Cenozoic); optimization for lower-end smartphones
Technologies	**Vuforia** - marker and image recognition; **mobile AR (Android)**; color recognition; AI narration for storytelling; **Google Cloud Text-to-Speech** and **Speech-to-Text**

3. Integration stack (cross-component)

er system integration documentation, shared or connected technologies include:

- **Unity Engine**
- **AR Foundation** + **Vuforia**
- **OpenXR** (VR)
- **Google Cloud Text-to-Speech**
- **LLM APIs** (narration / voice assistant)
- **Photon** (multiplayer networking)

4. Deployment architecture

4.1 Application split

Deliverable	Contents	Primary platforms
VR application	Component 1 + Component 2	**Meta Quest 3** (OpenXR)
AR application	Component 2 + Component 4	**Android** (primary); **Android and iOS** where Component 2's AR Foundation builds are released

4.2 VR application (single build)

- Modules included: Fossil excavation VR; multiplayer VR museum and jungle simulation.
- Shared technical base: Unity, OpenXR, Quest target hardware.
- Additional services: Photon (multiplayer); LLM integration for AI voice features as implemented in Component 4.
- Distribution: Meta Quest distribution channels appropriate to the release (e.g. App Lab, Release Channels, or institutional device deployment)

4.3 AR application (single build)

- Modules included: AI voice AR guide (QR, TTS, LLM narration); geological AR exploration (Vuforia, coloring-to-AR, quizzes).
- Shared technical base: Unity; Vuforia; AR Foundation where used by Component 2.
- Additional services: Google Cloud TTS; LLM APIs per Component 2 design.
- Distribution: Google Play (Android); Apple App Store (iOS) if the combined product ships for iPhone/iPad.

4.4 Backend and connectivity

- **AI:** Local or cloud deployment of LLM and TTS services (as configured per module).
- **Multiplayer:** Photon cloud service for Component 4 features inside the VR application.
- **Network:** Museum or venue Wi-Fi must support required API calls for TTS/LLM and Photon sessions where online features are used.

5. Hardware and software prerequisites

5.1 End users : VR application

- Headset: Meta Quest 2 or Quest 3
- Software: Build compiled for Quest Android/OpenXR stack

5.2 End users: AR application

- Android: Devices meeting ARCore and project minimum API requirements; Vuforia-compatible devices for marker/image tracking

5.3 Development and build

- Unity (team-aligned editor version; GeoQuest reference: 6000.2.6f1)
- Platform modules: Android
- SDKs: Vuforia, AR Foundation, OpenXR as per Package Manager configuration
- Accounts: Google Cloud (TTS); Photon; LLM provider APIs

6. Build, packaging, and versioning

- Version control: Single or multiple Unity projects merged into two deployable products; version numbers and changelogs maintained per VR app and AR app.
- Signing: Android keystore / iOS certificates and provisioning for store or enterprise distribution.
- Quest: Android signing and Meta submission requirements for the chosen channel.
- Configuration management: API keys and endpoints for TTS, LLM, and Photon supplied via secure configuration (not committed in plain text in public repositories).

7. Testing and quality assurance

Area	Scope
Functional	All four feature sets reachable in their respective combined app; QR, Vuforia targets, excavation tools, multiplayer sessions, voice/AI flows
Devices	Quest 3; representative Android devices (including budget-tier for Component 3); iOS devices if iOS is in release scope
Performance	Real-time rendering; AR frame rate on target phones; VR comfort and frame stability
Accessibility	Component 2 settings (text, speech speed, color-blind UI) where integrated into the AR app
Network	Photon under venue-like latency; TTS/LLM availability and failure behavior

8. Security, privacy, and operations

- API keys: Restricted credentials; rotation policy; separate dev/staging/production where applicable.
- Voice and AI: User awareness of voice/AI features; data handling consistent with LLM and cloud provider policies.
- Updates: Planned update path for app binaries and for Vuforia databases / content bundles if exhibits change.

9. Risks and dependencies

Risk / dependency	Description
Third-party services	Google Cloud TTS, LLM APIs, Photon require active accounts, quotas, and network access.
Platform policy	App Store, Play, and Meta policies must be satisfied for public release.
Hardware diversity	Android fragmentation affects AR performance; testing matrix must cover low-end devices referenced in Component 3.
Unity merge	Combining modules assumes compatible Unity editor, render pipeline, and package versions across components.

10. Adoption and rollout (project plan)

Aligned with the stated adoption approach:

- Pilot deployment in a museum setting
- Collect visitor and educator feedback
- Refine features from testing results
- Expand toward broader rollout

Target sectors: museums, science centers, educational institutions, cultural heritage sites.

11. Commercial and value summary

Bridging Realities turns conventional museum visits into active, memorable learning sessions. Instead of reading panels alone, visitors explore fossils in VR, walk through prehistoric settings with others, scan exhibits in AR, and hear explanations that can adapt to language and level. That shift supports stronger attention, retention, and emotional connection to science and history content.

Value to visitors and learning outcomes

- **Interactive learning** : Users discover content by doing (excavating, moving through spaces, scanning QR codes, coloring and revealing AR), which supports deeper understanding than passive viewing.
- **Immersive storytelling** : Geological eras, museum galleries, and jungle environments are shown in three dimensions and motion, making abstract timelines and ecosystems easier to grasp.
- **AI-driven personalization** : Narration and assistance can adjust to different languages, ages, and skill levels, widening who can benefit from the same installation.
- **Gamified engagement** : Quizzes, badges, challenges, and achievements encourage exploration and repeat use without replacing educational intent.

Commercial positioning

The platform is aimed at organizations that already invest in public education but want modern, repeatable experiences: **museums**, **science centers**, **schools and universities**, and **cultural heritage sites**. Hardware is familiar, smartphones and Meta Quest-class headsets, so entry barriers for venues are lower than custom installations that require rare or proprietary equipment.